

# XO Arena — Performance Trend

Auto-regenerated after every baseline run

## Contents

|                                       |          |
|---------------------------------------|----------|
| <b>XO Arena — Performance Trend</b>   | <b>1</b> |
| Targets (from Performance_Plan_v2.md) | 1        |
| Cold-page — Home, mobile + desktop    | 1        |
| Realtime + backend                    | 3        |
| Regressions detected                  | 4        |
| Coverage                              | 6        |

## XO Arena — Performance Trend

Generated: 2026-05-16T13:22:31.243Z

Auto-regenerated by perf/perf-summarize.js from every JSON in perf/baselines/. Wire into the tail of /stage and /promote to keep this fresh on every staging cut + prod promote.

### Targets (from Performance\_Plan\_v2.md)

- Mobile cold-anon Ready p50: **≤ 2000 ms**
- Desktop cold-anon Ready p50: **≤ 1000 ms**
- Mobile LCP p50: **≤ 1500 ms**
- SSE round-trip (POST move → SSE state) p50: **≤ 300 ms**
- Backend endpoint p95 (worst of measured): **≤ 200 ms**
- Mobile TBT p50: **≤ 100 ms**

Regressions flagged ☐ when a p95 grew >10% vs the previous same-env run.

### Cold-page — Home, mobile + desktop

| Date       | Version         | Env     | mob Ready        | dsk Ready                              | mob LCP         | mob TBT                               | img KB    | js KB      | regressions       |
|------------|-----------------|---------|------------------|----------------------------------------|-----------------|---------------------------------------|-----------|------------|-------------------|
| 2026-05-16 | 1.4.0-alpha-5.8 | staging | 1960ms<br>-5%    | 639ms -8%<br>↓                         | 1568ms<br>-5% ↓ | 38ms<br>+0%                           | 50<br>+0% | 517<br>+0% | —                 |
| 2026-05-16 | 1.4.0-alpha-5.8 | prod    | 1557ms<br>-20% ↓ | 670ms<br>+14% <input type="checkbox"/> | 272ms<br>-83% ↓ | 35ms<br>-3%                           | 50<br>+0% | 517<br>+0% | dsk Ready<br>+14% |
| 2026-05-16 | 1.4.0-alpha-5.8 | prod    | 1954ms<br>-4%    | 588ms<br>-23% ↓                        | 1580ms<br>-3%   | 36ms<br>-43% ↓                        | 50<br>+0% | 516<br>+0% | —                 |
| 2026-05-15 | 1.4.0-alpha-5.6 | staging | 2060ms<br>+1%    | 691ms<br>+2%                           | 1656ms<br>-1%   | 38ms<br>-12% ↓                        | 50<br>+0% | 516<br>+0% | —                 |
| 2026-05-15 | 1.4.0-alpha-5.6 | prod    | 2038ms<br>+0%    | 763ms<br>+9%                           | 1628ms<br>+0%   | 63ms<br>+62% <input type="checkbox"/> | 50<br>+0% | 516<br>+1% | mob TBT<br>+62%   |
| 2026-05-15 | 1.4.0-alpha-5.6 | staging | 2030ms<br>-4%    | 679ms<br>+1%                           | 1668ms<br>-3%   | 43ms<br>-46% ↓                        | 50<br>+0% | 516<br>+1% | —                 |
| 2026-05-15 | 1.4.0-alpha-5.6 | prod    | 2033ms<br>-4%    | 697ms<br>-17% ↓                        | 1620ms<br>-5%   | 39ms<br>+15% <input type="checkbox"/> | 50<br>+0% | 512<br>+1% | mob TBT<br>+15%   |
| 2026-05-15 | 1.4.0-alpha-5.6 | staging | 2104ms<br>+0%    | 673ms<br>-19% ↓                        | 1716ms<br>+1%   | 79ms<br>+80% <input type="checkbox"/> | 50<br>+0% | 512<br>+0% | mob TBT<br>+80%   |

| Date       | Version          | Env     | mob Ready        | dsk Ready       | mob LCP           | mob TBT        | img KB    | js KB      | regressions                                             |
|------------|------------------|---------|------------------|-----------------|-------------------|----------------|-----------|------------|---------------------------------------------------------|
| 2026-05-15 | 1.4.0-alpha-5.6  | staging | 2102ms<br>-3%    | 829ms -2%       | 1700ms<br>-6% ↓   | 44ms<br>+7%    | 50<br>+0% | 512<br>+0% | —                                                       |
| 2026-05-15 | 1.4.0-alpha-5.6  | staging | 2164ms<br>+8%    | 848ms<br>+20% □ | 1804ms<br>+12% □  | 41ms<br>+0%    | 50<br>+0% | 510<br>+1% | dsk Ready<br>+20%, mob<br>LCP +12%                      |
| 2026-05-14 | 1.4.0-alpha-5.1  | staging | 2013ms<br>+29% □ | 706ms<br>+3%    | 1604ms<br>+428% □ | 41ms<br>-20% ↓ | 50<br>+0% | 506<br>+0% | mob Ready<br>+29%, mob<br>LCP +428%                     |
| 2026-05-13 | 1.4.0-alpha-4.15 | prod    | 2121ms<br>+4%    | 838ms<br>+29% □ | 1704ms<br>+4%     | 34ms<br>-26% ↓ | 50<br>+0% | 506<br>+0% | dsk Ready<br>+29%                                       |
| 2026-05-13 | 1.4.0-alpha-4.15 | staging | 1556ms<br>+1%    | 688ms<br>+5%    | 304ms<br>+7%      | 51ms<br>+21% □ | 50<br>+0% | 506<br>+1% | mob TBT<br>+21%                                         |
| 2026-05-11 | 1.4.0-alpha-4.12 | prod    | 2039ms<br>-5%    | 648ms<br>-14% ↓ | 1636ms<br>-3%     | 46ms<br>+35% □ | 50<br>+0% | 504<br>+1% | mob TBT<br>+35%                                         |
| 2026-05-10 | 1.4.0-alpha-4.12 | staging | 1537ms<br>-6% ↓  | 653ms<br>-12% ↓ | 284ms<br>-82% ↓   | 42ms<br>+8%    | 50<br>+0% | 502<br>+0% | —                                                       |
| 2026-05-10 | 1.4.0-alpha-4.12 | staging | 1629ms<br>-25% ↓ | 743ms<br>-21% ↓ | 1588ms<br>-10% ↓  | 39ms<br>-3%    | 50<br>+0% | 500<br>+1% | —                                                       |
| 2026-05-09 | 1.4.0-alpha-4.9  | prod    | 2138ms<br>+1%    | 756ms<br>+2%    | 1684ms<br>+0%     | 34ms<br>-21% ↓ | 50<br>+0% | 497<br>+0% | —                                                       |
| 2026-05-09 | 1.4.0-alpha-4.9  | staging | 2175ms<br>+27% □ | 940ms<br>+9%    | 1764ms<br>+611% □ | 40ms<br>+5%    | 50<br>+0% | 497<br>+0% | mob Ready<br>+27%, mob<br>LCP +611%                     |
| 2026-05-09 | 1.4.0-alpha-4.9  | prod    | 2120ms<br>+31% □ | 738ms -3%       | 1680ms<br>+438% □ | 43ms<br>+23% □ | 50<br>+0% | 497<br>+0% | mob Ready<br>+31%, mob<br>LCP<br>+438%, mob<br>TBT +23% |
| 2026-05-09 | 1.4.0-alpha-4.9  | staging | 1706ms<br>+7%    | 865ms<br>+16% □ | 248ms<br>-19% ↓   | 38ms<br>-37% ↓ | 50<br>+0% | 497<br>+0% | dsk Ready<br>+16%                                       |
| 2026-05-06 | 1.4.0-alpha-4.7  | prod    | 1624ms<br>-9% ↓  | 763ms<br>-15% ↓ | 312ms<br>-22% ↓   | 35ms<br>-26% ↓ | 50<br>+0% | 497<br>+0% | —                                                       |
| 2026-05-06 | 1.4.0-alpha-4.7  | staging | 1601ms<br>-6% ↓  | 748ms<br>-20% ↓ | 308ms<br>-25% ↓   | 60ms<br>-21% ↓ | —<br>+0%  | 497<br>+0% | —                                                       |
| 2026-05-05 | 1.4.0-alpha-4.7  | staging | 1701ms<br>-3%    | 930ms<br>+4%    | 408ms<br>+2%      | 76ms<br>+29% □ | 50<br>+0% | 497<br>+0% | mob TBT<br>+29%                                         |
| 2026-05-05 | 1.4.0-alpha-4.7  | prod    | 1784ms<br>-14% ↓ | 893ms<br>+2%    | 400ms<br>-78% ↓   | 47ms<br>-53% ↓ | 50<br>+0% | 496<br>+0% | —                                                       |
| 2026-05-05 | 1.4.0-alpha-4.7  | staging | 1745ms<br>-5%    | 895ms -2%       | 400ms<br>-10% ↓   | 59ms<br>+48% □ | 50<br>+0% | 496<br>+0% | mob TBT<br>+48%                                         |
| 2026-05-05 | 1.4.0-alpha-4.7  | staging | 1833ms<br>+2%    | 910ms<br>+5%    | 444ms<br>+0%      | 40ms<br>+3%    | 50<br>+0% | 495<br>-0% | —                                                       |
| 2026-05-05 | 1.4.0-alpha-4.7  | staging | 1804ms<br>+3%    | 869ms<br>-10% ↓ | 444ms<br>+3%      | 39ms<br>+22% □ | 50<br>+0% | 496<br>+0% | mob TBT<br>+22%                                         |
| 2026-05-05 | 1.4.0-alpha-4.7  | staging | 1760ms<br>-14% ↓ | 970ms -3%       | 432ms<br>-76% ↓   | 32ms<br>-36% ↓ | 50<br>+0% | 496<br>+0% | —                                                       |
| 2026-05-05 | 1.4.0-alpha-4.7  | staging | 2041ms<br>-1%    | 998ms<br>+19% □ | 1792ms<br>-1%     | 50ms<br>-19% ↓ | 50<br>+0% | 495<br>-0% | dsk Ready<br>+19%                                       |
| 2026-05-05 | 1.4.0-alpha-4.7  | prod    | 2081ms<br>+3%    | 876ms<br>+16% □ | 1844ms<br>+3%     | 99ms<br>+87% □ | —<br>+0%  | 496<br>+0% | dsk Ready<br>+16%, mob<br>TBT +87%                      |
| 2026-05-04 | 1.4.0-alpha-4.0  | prod    | 2025ms<br>-1%    | 756ms -8%<br>↓  | 1792ms<br>-0%     | 53ms<br>+2%    | —<br>+0%  | 496<br>+0% | —                                                       |
| 2026-05-04 | 1.4.0-alpha-4.0  | prod    | 2040ms           | 819ms           | 1800ms            | 52ms           | —         | 496        | —                                                       |
| 2026-05-04 | 1.4.0-alpha-4.0  | staging | 2052ms           | 836ms           | 1816ms            | 62ms           | —         | 496        | —                                                       |

## Realtime + backend

| Date       | Version          | Env     | SSE p50          | SSE p95         | SSE fail | backend worst p95 | INP max p50 | note |
|------------|------------------|---------|------------------|-----------------|----------|-------------------|-------------|------|
| 2026-05-16 | 1.4.0-alpha-5.8  | staging | 279ms<br>+32% □  | 302ms +1%       | 0/20     | 85ms -38% ↓       | 32ms        | —    |
| 2026-05-16 | 1.4.0-alpha-5.8  | prod    | 230ms<br>-13% ↓  | 291ms -3%       | 0/20     | 119ms +143% □     | 32ms        | —    |
| 2026-05-16 | 1.4.0-alpha-5.8  | prod    | 265ms<br>+32% □  | 300ms -0%       | 0/20     | 49ms -89% ↓       | 32ms        | —    |
| 2026-05-15 | 1.4.0-alpha-5.6  | staging | 212ms<br>-18% ↓  | 298ms<br>-19% ↓ | 0/20     | 136ms -53% ↓      | 32ms        | —    |
| 2026-05-15 | 1.4.0-alpha-5.6  | prod    | 201ms<br>-9% ↓   | 301ms<br>+12% □ | 0/20     | 463ms +57% □      | 24ms        | —    |
| 2026-05-15 | 1.4.0-alpha-5.6  | staging | 259ms<br>+6%     | 366ms<br>+13% □ | 0/20     | 288ms +13% □      | 32ms        | —    |
| 2026-05-15 | 1.4.0-alpha-5.6  | prod    | 221ms<br>-22% ↓  | 269ms<br>-34% ↓ | 0/20     | 294ms -10% ↓      | 32ms        | —    |
| 2026-05-15 | 1.4.0-alpha-5.6  | staging | 244ms<br>-28% ↓  | 325ms<br>-27% ↓ | 0/20     | 254ms -13% ↓      | 32ms        | —    |
| 2026-05-15 | 1.4.0-alpha-5.6  | staging | 340ms<br>+19% □  | 443ms<br>+19% □ | 0/20     | 291ms +7%         | 32ms        | —    |
| 2026-05-15 | 1.4.0-alpha-5.6  | staging | 285ms<br>+17% □  | 372ms<br>+18% □ | 0/20     | 272ms -5%         | 32ms        | —    |
| 2026-05-14 | 1.4.0-alpha-5.1  | staging | 243ms<br>+4%     | 316ms +5%       | 0/20     | 286ms -6% ↓       | 32ms        | —    |
| 2026-05-13 | 1.4.0-alpha-4.15 | prod    | 284ms<br>+9%     | 405ms -3%       | 0/20     | 328ms +10% □      | 32ms        | —    |
| 2026-05-13 | 1.4.0-alpha-4.15 | staging | 233ms<br>-9% ↓   | 302ms -1%       | 0/20     | 304ms +14% □      | 32ms        | —    |
| 2026-05-11 | 1.4.0-alpha-4.12 | prod    | 260ms<br>-12% ↓  | 416ms<br>+19% □ | 0/20     | 297ms +115% □     | 40ms        | —    |
| 2026-05-10 | 1.4.0-alpha-4.12 | staging | 256ms<br>+2%     | 305ms +1%       | 0/20     | 267ms +56% □      | 40ms        | —    |
| 2026-05-10 | 1.4.0-alpha-4.12 | staging | 252ms<br>-18% ↓  | 301ms<br>-28% ↓ | 0/20     | 171ms -4%         | 32ms        | —    |
| 2026-05-09 | 1.4.0-alpha-4.9  | prod    | 294ms<br>+1%     | 349ms -1%       | 0/20     | 138ms -10% ↓      | 32ms        | —    |
| 2026-05-09 | 1.4.0-alpha-4.9  | staging | 308ms<br>+14% □  | 420ms +7%       | 0/20     | 178ms -4%         | 32ms        | —    |
| 2026-05-09 | 1.4.0-alpha-4.9  | prod    | 290ms<br>+6%     | 354ms<br>+22% □ | 0/20     | 153ms +1%         | 32ms        | —    |
| 2026-05-09 | 1.4.0-alpha-4.9  | staging | 271ms            | 394ms           | 0/20     | 186ms +14% □      | —           | —    |
| 2026-05-06 | 1.4.0-alpha-4.7  | prod    | 273ms<br>-14% ↓  | 289ms<br>-28% ↓ | 0/20     | 151ms -11% ↓      | 32ms        | —    |
| 2026-05-06 | 1.4.0-alpha-4.7  | staging | —                | —               | —        | 163ms -1%         | 24ms        | —    |
| 2026-05-05 | 1.4.0-alpha-4.7  | staging | 263ms<br>-17% ↓  | 358ms +1%       | 0/20     | 165ms +15% □      | 32ms        | —    |
| 2026-05-05 | 1.4.0-alpha-4.7  | prod    | 319ms<br>-45% ↓  | 403ms<br>-56% ↓ | 0/20     | 170ms +21% □      | 32ms        | —    |
| 2026-05-05 | 1.4.0-alpha-4.7  | staging | 317ms<br>+10% □  | 356ms<br>-22% ↓ | 0/20     | 144ms -5% ↓       | 40ms        | —    |
| 2026-05-05 | 1.4.0-alpha-4.7  | staging | 288ms<br>+1%     | 456ms +5%       | 0/20     | 152ms -45% ↓      | 32ms        | —    |
| 2026-05-05 | 1.4.0-alpha-4.7  | staging | 284ms<br>+4%     | 436ms<br>-52% ↓ | 0/20     | 274ms +11% □      | 32ms        | —    |
| 2026-05-05 | 1.4.0-alpha-4.7  | staging | 273ms<br>-66% ↓  | 905ms +0%       | 0/20     | 247ms +54% □      | 40ms        | —    |
| 2026-05-05 | 1.4.0-alpha-4.7  | staging | 806ms<br>+137% □ | 905ms +9%       | 0/20     | 160ms -18% ↓      | 40ms        | —    |

| Date       | Version         | Env     | SSE p50         | SSE p95          | SSE fail | backend worst p95 | INP max p50 | note                                     |
|------------|-----------------|---------|-----------------|------------------|----------|-------------------|-------------|------------------------------------------|
| 2026-05-05 | 1.4.0-alpha-4.7 | prod    | 577ms<br>+27%   | 926ms<br>-23% ↓  | 0/20     | 140ms +28% □      | 32ms        | recovery<br>from<br>17/20<br>bro-<br>ken |
| 2026-05-04 | 1.4.0-alpha-4.0 | prod    | 454ms<br>-38% ↓ | 1206ms<br>+13% □ | 17/20    | 109ms -54% ↓      | 32ms        | —                                        |
| 2026-05-04 | 1.4.0-alpha-4.0 | prod    | 731ms           | 1065ms           | 10/20    | 239ms             | 32ms        | —                                        |
| 2026-05-04 | 1.4.0-alpha-4.0 | staging | 340ms           | 829ms            | 0/20     | 196ms             | 32ms        | —                                        |

## Regressions detected

### 2026-05-16 — staging (v1.4.0-alpha-5.8)

- SSE p50 212 → 279 (+32%)

### 2026-05-16 — prod (v1.4.0-alpha-5.8)

- dsk Ready 588 → 670 (+14%)
- backend p95 49 → 119 (+143%)

### 2026-05-16 — prod (v1.4.0-alpha-5.8)

- SSE p50 201 → 265 (+32%)

### 2026-05-15 — prod (v1.4.0-alpha-5.6)

- SSE p95 269 → 301 (+12%)
- mob TBT 39 → 63 (+62%)
- backend p95 294 → 463 (+57%)

### 2026-05-15 — staging (v1.4.0-alpha-5.6)

- SSE p95 325 → 366 (+13%)
- backend p95 254 → 288 (+13%)

### 2026-05-15 — prod (v1.4.0-alpha-5.6)

- mob TBT 34 → 39 (+15%)

### 2026-05-15 — staging (v1.4.0-alpha-5.6)

- mob TBT 44 → 79 (+80%)

### 2026-05-15 — staging (v1.4.0-alpha-5.6)

- SSE p50 285 → 340 (+19%)
- SSE p95 372 → 443 (+19%)

### 2026-05-15 — staging (v1.4.0-alpha-5.6)

- dsk Ready 706 → 848 (+20%)
- mob LCP 1604 → 1804 (+12%)
- SSE p50 243 → 285 (+17%)
- SSE p95 316 → 372 (+18%)

### 2026-05-14 — staging (v1.4.0-alpha-5.1)

- mob Ready 1556 → 2013 (+29%)
- mob LCP 304 → 1604 (+428%)

**2026-05-13 — prod (v1.4.0-alpha-4.15)**

- **dsk Ready** 648 → 838 (+29%)
- **backend p95** 297 → 328 (+10%)

**2026-05-13 — staging (v1.4.0-alpha-4.15)**

- **mob TBT** 42 → 51 (+21%)
- **backend p95** 267 → 304 (+14%)

**2026-05-11 — prod (v1.4.0-alpha-4.12)**

- **SSE p95** 349 → 416 (+19%)
- **mob TBT** 34 → 46 (+35%)
- **backend p95** 138 → 297 (+115%)

**2026-05-10 — staging (v1.4.0-alpha-4.12)**

- **backend p95** 171 → 267 (+56%)

**2026-05-09 — staging (v1.4.0-alpha-4.9)**

- **mob Ready** 1706 → 2175 (+27%)
- **mob LCP** 248 → 1764 (+611%)
- **SSE p50** 271 → 308 (+14%)

**2026-05-09 — prod (v1.4.0-alpha-4.9)**

- **mob Ready** 1624 → 2120 (+31%)
- **mob LCP** 312 → 1680 (+438%)
- **SSE p95** 289 → 354 (+22%)
- **mob TBT** 35 → 43 (+23%)

**2026-05-09 — staging (v1.4.0-alpha-4.9)**

- **dsk Ready** 748 → 865 (+16%)
- **backend p95** 163 → 186 (+14%)

**2026-05-05 — staging (v1.4.0-alpha-4.7)**

- **mob TBT** 59 → 76 (+29%)
- **backend p95** 144 → 165 (+15%)

**2026-05-05 — prod (v1.4.0-alpha-4.7)**

- **backend p95** 140 → 170 (+21%)

**2026-05-05 — staging (v1.4.0-alpha-4.7)**

- **SSE p50** 288 → 317 (+10%)
- **mob TBT** 40 → 59 (+48%)

**2026-05-05 — staging (v1.4.0-alpha-4.7)**

- **mob TBT** 32 → 39 (+22%)
- **backend p95** 247 → 274 (+11%)

**2026-05-05 — staging (v1.4.0-alpha-4.7)**

- **backend p95** 160 → 247 (+54%)

#### 2026-05-05 — staging (v1.4.0-alpha-4.7)

- **dsk Ready** 836 → 998 (+19%)
- **SSE p50** 340 → 806 (+137%)

#### 2026-05-05 — prod (v1.4.0-alpha-4.7)

- **dsk Ready** 756 → 876 (+16%)
- **mob TBT** 53 → 99 (+87%)
- **backend p95** 109 → 140 (+28%)

#### 2026-05-04 — prod (v1.4.0-alpha-4.0)

- **SSE p95** 1065 → 1206 (+13%)

### Coverage

| Run start (UTC)  | Env     | Scripts present                                                                    |
|------------------|---------|------------------------------------------------------------------------------------|
| 2026-05-16 13:12 | staging | backend-p95, inp, longtasks, perf, playvsbot, playvsbot-mobile, sse-rtt, waterfall |
| 2026-05-16 11:45 | prod    | backend-p95, inp, longtasks, perf, playvsbot, playvsbot-mobile, sse-rtt, waterfall |
| 2026-05-16 00:11 | prod    | backend-p95, inp, longtasks, perf, playvsbot, sse-rtt, waterfall                   |
| 2026-05-15 20:44 | staging | backend-p95, inp, longtasks, perf, playvsbot, sse-rtt, waterfall                   |
| 2026-05-15 19:22 | prod    | backend-p95, inp, longtasks, perf, playvsbot, sse-rtt, waterfall                   |
| 2026-05-15 17:41 | staging | backend-p95, inp, longtasks, perf, sse-rtt, waterfall                              |
| 2026-05-15 14:28 | prod    | backend-p95, inp, longtasks, perf, sse-rtt, waterfall                              |
| 2026-05-15 13:51 | staging | backend-p95, inp, longtasks, perf, sse-rtt, waterfall                              |
| 2026-05-15 10:50 | staging | backend-p95, inp, longtasks, perf, sse-rtt, waterfall                              |
| 2026-05-15 01:40 | staging | backend-p95, inp, longtasks, perf, sse-rtt, waterfall                              |
| 2026-05-14 15:36 | staging | backend-p95, inp, longtasks, perf, sse-rtt, waterfall                              |
| 2026-05-13 22:46 | prod    | backend-p95, inp, longtasks, perf, sse-rtt, waterfall                              |